VIVA PRESENTATION & STUDY GUIDE (HINGLISH)

AI-Based Patient Appointment System

Milestone 1, Milestone 2 aur Milestone 3 ka Complete Master Explanation Guide

PROJECT DOMAIN
Healthcare AI & Full-Stack Web Application

CORE TECH STACK
React 18 + FastAPI + Python NLP + SQLite + Google OAuth

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B.Tech Information Technology (B.Tech IT)
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1. Project Ka Main Idea (30-Second Intro for Viva)

> **Examiner ke aage bolne ke liye 30-Second Elevator Pitch:**
> *"Good morning Sir/Ma'am. Hamara project hai **AI-Based Patient Appointment System**. Aaj kal Practo ya Zocdoc jaise apps me patient ko khud decide karna padta hai ki kis doctor ke paas jaana hai (self-diagnosis), jisse galat specialist book ho jaata hai. Aur Ada Health jaise symptom checkers sirf bimari diagnose karte hain par doctor ka appointment book nahi karate. Hamara system dono ko jodta hai — jisme **24/7 AI Chatbot** patient ke symptoms analyze karta hai, **Emergency Red-Flags** detect karta hai, sahi **Doctor Specialist** recommend karta hai, **Conflict-Free Real-Time Booking** karata hai, aur Doctor ko pehle se **AI Clinical Summary** bhejta hai."*

12+	< 1.5 sec	100%
DOCTOR SPECIALTIES	AI TRIAGE SPEED	NO DOUBLE-BOOKING

Hamare Project ki 4 Sabse Badi Khasiyat (Core USPs)

1. **AI Symptom Triage:** Mareez ko apni bhasha me symptoms likhne hain, AI turant samajh leta hai.
2. **Emergency Bypass:** Agar patient ko Heart Attack ya Stroke ke symptoms hain (jaise chest pain, slurred speech), toh routine booking rok kar turant **Emergency Red Alert** deta hai.
3. **Right Doctor Recommendation:** Symptoms ke hisab se exact specialist (Cardiologist, Neurologist, Pulmonologist, etc.) filter karke deta hai.
4. **Doctor Pre-Consultation Summary:** Doctor ko patient aane se pehle summary mil jaati hai, jisse OPD me 10 minute ka time bachta hai.

2. Teeno Milestones Me Kya-Kya Hua Hai? (Summary)

Milestone	Kyu Kiya Gaya? (Objective)	Isme Kya Kaam Kiya? (Work Done)	Output / Deliverables
Milestone 1 MARKET STUDY & REQ	Existing systems ko study karna, gaps dhundhna aur project ke requirements tay karna.	• Practo, Zocdoc, Ada Health, Babylon Health ka comparative study kiya. • 3 main market gaps identify kiye. • 5 Functional (FR) aur 4 Non-Functional (NFR) requirements banaye.	• Comparative Matrix • Problem Statement • SRS Document
Milestone 2 DATASET & AI ENGINE	Medical dataset banana, clean karna, EDA karna aur AI Recommendation Engine code karna.	• Columbia University, Mayo Clinic, ICD-10 se medical dataset banaya. • 6-stage data preprocessing pipeline banayi (cleaning, tokenizing). • 2-Tier AI Triage algorithm code kiya (Emergency Check + Jaccard Similarity).	• `symptom_specialist_dataset.csv` • `ai_engine.py` module • EDA Analysis
Milestone 3 SYSTEM ARCHITECTURE	Full-stack system architecture, 8 functional modules, database ER schema aur auth flows design karna.	• 4-tier decoupled architecture banayi. • 8 Core modules define kiye (Auth, Chatbot, AI, Doctor Directory, Booking, etc.). • 3NF relational DB schema banaya (SQLAlchemy). • Google OAuth 2.0 (GIS) + JWT token security design ki.	• Architecture Diagram • Relational ER Schema • REST API Endpoints • Security & Disclaimers

3. Milestone 1 Detail: Existing Systems vs Hamara System

3.1 Existing Platforms me Kya Kami Thi? (Gap Analysis)

- **Practo / Zocdoc:** Doctor directory achhi hai, booking bhi hoti hai, par **AI Symptom Checker nahi hai**. Patient ko khud guess karna padta hai ki kis specialist ke paas jaaye.
- **Ada Health / Buoy Health:** AI symptom checker bohot achha hai, par **Doctor appointment booking nahi hai**. Diagnosis dekh kar user ko doosre app me jaana padta hai.
- **Hamara System:** Dono ko ek single platform me seamlessly jod diya hai!

3.2 Functional Requirements (FR) & Non-Functional Requirements (NFR)

Functional Requirements (FR)

- **FR-1:** User Registration & Google Login (JWT).
- **FR-2:** AI Conversational Chatbot Drawer.
- **FR-3:** Automated Doctor Specialist Mapping.
- **FR-4:** Real-time Conflict-free Slot Booking.
- **FR-5:** Dual Dashboards (Patient + Doctor).

Non-Functional Requirements (NFR)

- **NFR-1 Security:** AES-256 data at rest, TLS 1.3 in transit, Bcrypt password hashing.
- **NFR-2 Performance:** AI triage latency < 1.5s, API booking < 500ms.
- **NFR-3 Usability:** Responsive SPA UI across mobile & desktop.
- **NFR-4 Safety:** Medical disclaimer + ER emergency redirection.

4. Milestone 2 Detail: Dataset, Preprocessing & AI Engine

4.1 Data Cleaning Pipeline (6 Steps)

```
[Raw Clinical Data / User Input Text]
       |
       ▼  Step 1: UTF-8 Encoding Sanitization
       ▼  Step 2: Null / Empty Records Removal
       ▼  Step 3: Deduplication (Duplicate symptoms hataye)
       ▼  Step 4: Clinical NLP Tokenization (Lowercase + Regex \w+)
       ▼  Step 5: Medical Synonym Harmonization ("dyspnea" → "shortness of breath")
       ▼  Step 6: Urgency Stratification (Emergency / High / Medium / Low)
       |
[Clean Knowledge Base in backend/data/symptom_specialist_dataset.csv]
```

4.2 AI Algorithm Kaise Kaam Karta Hai? (Two-Tier Engine)

Tier 1: Emergency Red-Flag Classifier (Pehle Check)

Emergency keywords: `["chest pain", "shortness of breath", "slurred speech", "numbness in face", "arm pain", "stroke", "heart attack", "unconscious", "difficulty breathing"]`.

Agar user inme se kuch bhi bolta hai → System turant **EMERGENCY RED ALERT** deta hai aur normal booking cancel karke 112/108 helpline call karne ko kehta hai.

Tier 2: Tokenized Jaccard Similarity Matcher (Specialist Recommendation)
Agar emergency nahi hai, toh user input tokens aur dataset symptoms ke tokens ka overlap (Jaccard Similarity) nikalte hain: $$\text{Confidence} = \min\left(0.60 + (\text{Similarity} \times 0.35),\, 0.95\right)$$ **Safe Default Routing:** Agar similarity 15% se kam ho, toh safe side rehkar **General Physician** ko recommend kiya jaata hai (70% confidence).

5. Milestone 3 Detail: System Architecture, Modules & Tech Stack

5.1 Humne Kaunsi Technologies Use Ki Hain Aur Kyu?

Technology	Stack Layer	Kyu Use Kiya? (Viva Reason)
React 18 + Vite	Frontend UI	Bohot fast load time, reusable components (Chatbot Drawer, Slot Modal), smooth glassmorphism design.
FastAPI (Python)	Backend API	Asynchronous (`asyncio`) hone ki wajah se super-fast hai, Pydantic type safety deta hai, aur Swagger docs auto-generate karta hai.
Python NLP Engine	AI Triage	Fast execution (< 50ms), zero external API cost/failure, deterministic emergency rules.
SQLAlchemy + SQLite	Database & ORM	Relational integrity (ACID transactions) taaki do log ek hi slot ko double book na kar sakein.
Google OAuth 2.0 (GIS) + JWT	Authentication	One-click secure Google Sign-In, stateless cryptographic JWT session tokens, Bcrypt password hashing.

5.2 Project ke 8 Subsystem Modules

1. Auth & Identity Service
Google OAuth 2.0 GIS token verification, OTP verification, Bcrypt password hashing, JWT session tokens.

2. AI Chatbot Drawer
Interactive drawer jo right side se slide-in hota hai, conversational symptom intake ke liye.

3. Clinical Assessment Engine
User ke text se disease aur urgency calculate karta hai, emergency detect karta hai.

4. Doctor Recommendation Service
Diagnosed specialty ke hisab se verified doctors filter karke show karta hai.

5. Doctor Schedule Management
30-minute ke consultation slots maintain karta hai, real-time availability check karta hai.

6. Appointment Booking Engine
Atomic transaction booking karta hai taaki double-booking kabhi na ho sake.

7. Patient Health Records
Patient ki past booking history, reported symptoms aur AI diagnosis save rehta hai.

8. Doctor Clinical Dashboard
Doctor ko unke daily patients ki list aur AI-generated pre-consultation summary dikhata hai.

5.3 Double-Booking Prevention Kaise Hoti Hai?

Database me `(doctor_id, date, slot)` par **Unique Constraint** laga hai aur booking logic **Atomic Transaction** me chalti hai. Agar do patients ek hi waqt par same slot click karein, toh pehle wale ka book ho jaata hai aur doosre ko backend se **HTTP 409 Conflict** error milta hai, jisse UI me wo slot instantly disable ho jaata hai.

6. Top 10 Viva Questions & Answers (Examiner ke Saamne Bolne ke Liye)

🎯 Viva Question: Q1: Aapke project ka main objective kya hai?

Answer: Hamara main objective hai ek aisa intelligent healthcare system banana jo patient ke symptoms sunkar AI ki madad se bimari aur emergency detect kare, sahi specialist doctor recommend kare, real-time me bina slot conflict ke booking karaye, aur doctor ko consultation se pehle ek structured clinical summary provide kare.

🎯 Viva Question: Q2: Milestone 1, Milestone 2 aur Milestone 3 me kya difference hai?

Answer:

- **Milestone 1:** Market study (Practo vs Ada Health), gap identification, aur Functional / Non-Functional Requirements define kiye.
- **Milestone 2:** Medical datasets collect kiye, 6-stage data preprocessing ki, aur 2-Tier AI Triage recommendation engine code kiya.
- **Milestone 3:** Full system architecture banayi, 8 functional modules design kiye, relational ER database banaya, aur Google OAuth 2.0 / JWT security integrate ki.

🎯 Viva Question: Q3: Backend me Django ya Flask ke badle FastAPI kyu use kiya?

Answer: FastAPI modern asynchronous framework hai jo Python ke `asyncio` par chalta hai. Ye Flask aur Django se 3× fast hai, isme Pydantic ki vajah se automatic request validation milti hai, aur API test karne ke liye Swagger UI documentation automatically ban jaati hai.

🎯 Viva Question: Q4: Agar patient ko heart attack ke symptoms hain toh AI kya karega?

Answer: Hamara **Tier 1 Emergency Classifier** emergency keywords (jaise chest pain, left arm pain, breathlessness) ko detect karte hi normal clinic booking ko bypass kar deta hai aur screen par **Emergency Red Alert** dikhakar patient ko turant 112/108 helpline ya nearest hospital ER jaane ka instruction deta hai.

🎯 Viva Question: Q5: Agar patient ke symptoms kisi bhi specialist se match na karein toh kya hoga?

Answer: System me ek **Safe Default Fallback** mechanism hai. Agar similarity score threshold (≤ 15%) se kam rehta hai, toh AI patient ko safely **General Physician** (Internal Medicine) recommend karta hai, jo standard medical protocol hai.

🎯 Viva Question: Q6: Google Sign-In authentication kaise kaam karta hai?

Answer: Hum Google Identity Services (GIS) use karte hain. User jab Google button par click karta hai, toh Google frontend ko ek signed ID Token deta hai. Backend is token ko Google ke servers se verify karta hai, user ka profile save karta hai aur secure JWT session token return karta hai.

🎯 Viva Question: Q7: AI ke medical suggestions me legal ya safety risk ko kaise handle kiya hai?

Answer:

1. Har assessment ke saath statutory disclaimer aata hai ki ye preliminary AI triage hai, final diagnosis nahi.
2. **Doctor-in-the-Loop:** AI sirf doctor ke liye pre-consultation summary draft karta hai, final check aur prescription doctor hi karta hai.
3. Critical emergency me instant ER redirect hota hai.

🎯 Viva Question: Q8: Doctor ko is system se kya fayda hota hai?

Answer: Doctor ko unke dashboard par har patient ka pre-consultation AI summary milta hai jisme reported symptoms, duration aur preliminary diagnosis pehle se likha hota hai. Isse doctor ka OPD me 5–10 minute ka repetitive time bachta hai aur wo seedha treatment par focus kar sakte hain.